

Sign Reversal of the DC-Sweep Capacitance–Voltage Hysteresis Window in $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ Ferroelectric Transistors

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September 8, 2026

Abstract

Memory-window collapse in HfO_2 ferroelectric transistors is commonly attributed to ferroelectric fatigue. We show that the DC-sweep C – V hysteresis window of $\text{TaN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2(10\text{ nm})/\text{SiO}_2(0.85\text{ nm})/\text{p-Si}$ transistors does not merely close but *reverses sign*, becoming clockwise beyond 5×10^2 – 2×10^3 cycles. The reversal is reproduced on five independent devices, is unchanged when the measurement integration time is increased ten-fold, and is present at every frequency from 10 kHz to 1 MHz and every sweep amplitude from $\pm 1.5\text{ V}$ to $\pm 3\text{ V}$. Transfer characteristics confirm it: ΔV_{th} falls from 0.581 V to -0.015 V at 10^4 cycles, independently of the extraction criterion. Over the same range the remanent polarization *increases*, rising 45% by 10^2 cycles and remaining above its pristine value at 10^4 . The window efficiency consequently falls from $\sim 16\%$ to -4% . Interfacial charge first compensates, then over-compensates, the polarization field at the channel.

Index Terms— Ferroelectric field-effect transistor, hafnium oxide, endurance, hysteresis window, charge trapping, capacitance–voltage, interfacial layer, PUND.

1 Introduction

Ferroelectricity in doped HfO_2 [2, 1] made a CMOS-compatible ferroelectric field-effect transistor (FeFET) practical, and integrated FeFET technologies now exist at the 28 nm node and below [13, 14]. Endurance remains the principal obstacle: the memory window typically collapses between 10^4 and 10^5 write cycles [18, 33, 21], and the gap persists across silicon- and zirconium-doped stacks alike [8, 27].

The default explanation is ferroelectric fatigue [25]. An alternative attributes the collapse to charge injected into

the ferroelectric/interfacial-layer region, which screens the polarization from the channel without destroying it [32, 10, 30, 20]. Charge trapping, trap generation and interfacial-layer effects are established features of FeFET operation rather than mechanisms identified here [29, 17, 24]. The two mechanisms imply opposite remedies — ferroelectric materials engineering [16] versus interface engineering [28, 3, 22] — so distinguishing them matters. Reported evidence supports both, including oxygen-vacancy redistribution during cycling [15, 34, 11].

Two of our own recent studies bear directly on this work and we state the relationship explicitly. In [7] we showed that ferroelectric polarization survives after the FeFET memory window collapses. In [5] we extracted an apparent channel-coupled charge density from the threshold-voltage shift, defined a polarization-to-channel coupling factor by comparison with the switching charge, and reported its decrease from 3.5% to 0.9% at 10^4 cycles. Both establish that the coupling between polarization and channel degrades while polarization persists; we do not claim that observation as new. A companion preprint reports the stability of the gate-stack polarization response in degraded devices [6].

Three findings in the present work are not contained in those studies. First, the hysteresis window does not merely approach zero — it becomes *negative*. A coupling factor is non-negative by construction, so the regime in which interfacial charge *over-compensates* rather than merely screens the polarization field cannot be expressed in that framework. Beyond the crossing, a positive gate sweep leaves the transistor harder to turn on, the opposite of its pristine behavior. Second, the reversal is established on five independent devices, across three decades of measurement frequency, four sweep amplitudes and a ten-fold change in integration time. Third, it is resolved from the gate-stack C – V alone, and independently confirmed by transfer characteristics.

2 Device Fabrication and Measurement

2.1 Gate Stack

The device is a p-Si channel FeFET with a $\text{TaN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2(10\text{ nm})/\text{SiO}_2(0.85\text{ nm})$ gate stack

^{*}This work was partly funded by the National Science and Technology Council (NSTC), Taiwan, under grants 115-2221-E-007-144, by the Taiwan Semiconductor Research Institute (TSRI) grant JDP114-Y1-002, and by PSMC-115A0330.

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[Fig. 1(a)]. The chemical SiO_2 interfacial layer (IL) is unavoidable on a silicon channel and is the region of interest. The insulator stack forms a series capacitive divider,

$$\frac{V_{\text{HZO}}}{V_{\text{ins}}} = \frac{C_{\text{IL}}}{C_{\text{IL}} + C_{\text{HZO}}}, \quad (1)$$

where $V_{\text{ins}} = V_{\text{HZO}} + V_{\text{IL}}$. With $\epsilon_{\text{HZO}} = 30$ and $\epsilon_{\text{IL}} = 3.9$, $C_{\text{HZO}} = 2.66 \mu\text{F cm}^{-2}$ and $C_{\text{IL}} = 4.06 \mu\text{F cm}^{-2}$, giving 0.60. Because the semiconductor capacitance is in series and neglected in (1), 0.60 is an upper bound.

2.2 Devices and Data Provenance

Five nominally identical devices from the same die and process are reported, denoted device-1 to device-3 and device-6, device-7. Device-1 is the device of the original submission; the others were not previously measured. All were cycled under identical conditions.

Two measurement types were performed on single devices and we state this explicitly. Polarization (PUND) was measured on one device of the die; because polarization behavior is uniform across devices of this process [7, 5], the resulting $2P_r(N)$ is used as a *common reference* against which the window of every device is compared, rather than as a per-device quantity. Transfer characteristics were measured on a single device. Neither is presented as a device-to-device reproducibility result; that role is carried by the C - V data on five devices.

2.3 Polarization Measurement

Switching-current and P - E measurements used triangular waveforms with a half-period of 500 μs . Polarization was evaluated by positive-up-negative-down (PUND) measurement:

$$Q_{\text{sw}}^+ = \int [I_P - I_U] dt, \quad Q_{\text{sw}}^- = \int [I_N - I_D] dt, \quad (2)$$

where P , N denote switching and U , D nonswitching pulses, and

$$2P_r = \frac{|Q_{\text{sw}}^+| + |Q_{\text{sw}}^-|}{2A}. \quad (3)$$

Coercive voltages were taken from the P - V zero crossings, and

$$V_{\text{imp}} = \frac{V_c^+ + V_c^-}{2}, \quad E_c = \frac{|V_c^+| + |V_c^-|}{2d}. \quad (4)$$

For the pristine device $V_c = 2.08 \text{ V}$, which with (1) gives $E_c \approx 1.25 \text{ MV cm}^{-1}$, consistent with reported values [31, 9].

2.4 Cycling, Capacitance and Transfer Measurement

Cycling used $\pm 5 \text{ V}$, 50 μs pulses. C - V was measured on a Keysight B1500A CMU in C_p - R_p mode with a 20 mV AC

level. Devices 6 and 7 were characterized at seven frequencies (1 kHz–1 MHz) and four sweep amplitudes ($\pm 1.5 \text{ V}$ – $\pm 3 \text{ V}$) at each checkpoint; devices 2 and 3 at 100 kHz and $\pm 3 \text{ V}$ with two integration settings. Source, drain and body were tied so that the inversion branch is supplied with minority carriers.

Two features of the capacitance measurement require explicit statement. The sweep begins at the negative rail and integrates for one power-line cycle per point, so a full sweep takes $\sim 1 \text{ s}$ — four orders of magnitude slower than the write pulse. Consequently the pulsed state does not survive to mid-sweep, and C - V curves acquired after program and after erase are indistinguishable. The reported quantity is therefore the *DC-sweep hysteresis window*

$$\Delta V_{\text{hyst}} = V_{\text{fwd}} - V_{\text{rev}} \Big|_{C=C_{\min} + \frac{1}{2}(C_{\max} - C_{\min})}. \quad (5)$$

It is not the retained state of a pulse-written bit, and no claim about stored information is made from it alone; the corresponding transistor quantity is measured separately in Section III-E. The sign convention is fixed throughout: $\Delta V_{\text{hyst}} > 0$ denotes a counter-clockwise loop (the ferroelectric sense); $\Delta V_{\text{hyst}} < 0$ a clockwise loop (the electron-trapping sense).

Transfer characteristics were measured with an SPGU write ($\pm 5 \text{ V}$, 50 μs) and an SMU read ($V_D = 0.1 \text{ V}$, V_G swept -1.5 to 2.5 V). V_{th} was extracted at a constant current of 1 μA . Because the gate must be switched between the pulse and read configurations through a switching matrix, the read delay was 60 s, determined from measurement timestamps; all terminals were held at 0 V during that interval and each read was preceded by a fresh write.

2.5 Integration-Time Control

To test whether the reported hysteresis is affected by charge leaking during the ADC integration window, the cycling series was repeated on devices 2 and 3 with the integration coefficient increased from 1 to 10 power-line cycles, i.e. 16.7 ms to 167 ms per point, all else unchanged.

2.6 Series-Resistance Correction and Screening

A series resistance was extracted from the Nicolian–Brews strong-accumulation expression [19, 26]

$$R_s = \frac{G_{ma}}{G_{ma}^2 + \omega^2 C_{ma}^2}, \quad (6)$$

and capacitances corrected using

$$C_c = \frac{(G^2 + \omega^2 C^2)C}{a^2 + \omega^2 C^2}, \quad a = G - (G^2 + \omega^2 C^2)R_s. \quad (7)$$

Sweeps with median loss tangent $D = 1/\omega R_p C_p$ above the stated screen were excluded from window statistics.

2.7 Figures of Merit

An unscreened remanent polarization would produce a window $2P_r/C_{\text{IL}}$. We define the window efficiency

$$\eta = \frac{\Delta V_{\text{hyst}}}{2P_r/C_{\text{IL}}}, \quad (8)$$

using the common reference $2P_r(N)$ of Section II-B. Where a window deficit is attributed to interfacial charge, an *equivalent* areal density follows from

$$\Delta N_{\text{eq}} = \frac{C_{\text{IL}} \Delta V}{q} \approx \Delta V \times 2.5 \times 10^{13} \text{ cm}^{-2} \text{ V}^{-1}. \quad (9)$$

Equation (9) assumes a sheet charge at the interfacial layer and an unchanged C_{IL} ; it is an equivalent density inferred from a voltage shift, not a directly extracted interface-trap density, and is denoted N_{eq} for that reason.

3 Results and Discussion

3.1 The Ferroelectric Remains Intact to 10^4 Cycles

Fig. 2 shows the polarization response. Field cycling produces a pronounced wake-up, consistent with oxygen-vacancy redistribution and domain-wall de-pinning [4, 23]: $2P_r$ rises from $20.2 \mu\text{C cm}^{-2}$ pristine to $29.2 \mu\text{C cm}^{-2}$ at 10^2 cycles, a 45% increase, and the positive switching peak nearly doubles. Both remain at or above their pristine values through 10^4 . V_c is constant to within 4% at every verified checkpoint through 10^4 (Table 3).

Wake-up and charge trapping proceed simultaneously rather than sequentially: devices 2 and 3 show wake-up in the hysteresis window itself, rising from 0.822 V to 0.920 V and from 0.804 V to 0.968 V between pristine and one cycle before collapsing.

Degradation appears only at 10^5 cycles, where $2P_r$ falls to $18.6 \mu\text{C cm}^{-2}$, the peaks broaden, and the non-switching current rises. V_c *decreases* there, from 2.08 V to 1.94 V — not a domain-pinning signature — consistent with a leakier, electrically softened film [12]. We therefore do not claim that ferroelectric fatigue is absent. The ferroelectric does degrade, approximately two decades after the hysteresis window has collapsed, so loss of switchable polarization alone cannot account for the early collapse.

The imprint voltage drifts monotonically from -48 mV pristine to 9 mV at 10^4 , crossing zero between 10^3 and 10^4 , consistent with accumulation of a negative charge sheet on the channel side of the ferroelectric [9].

3.2 The Window Collapses and Reverses Sign

Fig. 3 shows the C - V evolution. The separation between forward and reverse branches narrows monotonically, closes near 10^3 cycles, and reopens at 10^4 with the branches exchanged.

Fig. 4(a) shows ΔV_{hyst} for five independent devices. All exhibit a counter-clockwise window when fresh and a clockwise window by 10^4 cycles; interpolated zero crossings are 503, 871, 1117, 2136 and 2422 cycles [Fig. 4(b)], a spread of a factor of five. At 10^4 the window is -0.15 to -0.28 V across all five (Table 1). Device-1 has a smaller pristine window (0.363 V) than the other four (0.79 – 0.82 V), reflecting die-to-die variation; the sign reversal is nonetheless common to all.

The window measured on the inversion flank — the flank corresponding to V_{th} — collapses and inverts on the same schedule [Fig. 4(c)], excluding an artifact of the extraction point, and is consistently larger than the flatband-flank window, indicating that the loop stretches rather than translating rigidly. A rigid shift, as produced by a fixed charge sheet, would displace both flanks equally. The reversal is also present at every sweep amplitude [Fig. 4(d)]; we note that the 50%-of-range extraction criterion is not directly comparable between amplitudes and therefore report only the persistence of the sign change, not a trend in magnitude.

3.3 The Reversal Is Not a Measurement Artifact

Three independent checks are shown in Fig. 5. First, ΔV_{hyst} is suppressed at low frequency and saturates above $\sim 100 \text{ kHz}$, following

$$\Delta V_{\text{hyst}}(f) = \Delta V_{\infty} - \frac{\delta V}{1 + (f/f_c)^2}, \quad (10)$$

with $f_c = 25.6 \text{ kHz}$ (device-6) and 16.8 kHz (device-7) in the pristine state, both consistent with the 22 – 34 kHz obtained on device-1 [Fig. 5(b)]. Fits at 10^2 cycles use only four points on a partly collapsed window and are shown as open symbols; we draw no conclusion from them. The reversal itself is present at all seven measurement frequencies from 1 kHz to 1 MHz .

Second, increasing the ADC integration ten-fold changes ΔV_{hyst} by 15 – 22% but does not change its sign at any checkpoint through 10^5 on either device [Fig. 5(c)]. In the reversed regime the magnitude moves *toward* zero at longer integration. Charge leaking during the integration window would produce the opposite dependence, since a longer integration would deepen an artifact-driven negative window. The observed trend is therefore inconsistent with that mechanism.

Third, the reversal is present in uncorrected C_p data: of the sweeps at 10^4 cycles on device-1 for which the flank crossing is defined without any series correction, 25 of 27 are negative, spanning five frequencies, four amplitudes and both pulse states.

3.4 Gate-Stack Controls

Fig. 6 shows the accumulation capacitance, loss spectra and conduction for devices 6 and 7. The validity of the

correction of (7) is checkpoint-dependent, and we restrict the claim accordingly: a physically meaningful corrected capacitance is obtained through 10^3 cycles on device-1 and through 10^2 on device-2, while at 10^4 cycles $\omega R_s C \approx 0.50$ and the correction is ill-conditioned. No corrected value is quoted there, and the claim of an unchanged capacitance is not extended to the checkpoint at which the sign reversal occurs.

Within the valid range the result is nonetheless decisive. On device-2, over $N = 1$ to 10^2 cycles, $C_{\text{acc}} = 184.2 \pm 1.6 \text{ pF}$ ($\pm 0.9\%$) with the correction below 1.9% at every point and $D = 0.12\text{--}0.15$, while ΔV_{hyst} falls from 0.920 V to 0.331 V — 64% of the collapse. The stack capacitance is therefore constant to better than 1%, on essentially uncorrected data, over the range in which the window loses most of its magnitude.

3.5 Transistor-Level Confirmation

Fig. 8 shows transfer characteristics measured after program and erase at each checkpoint. Defining $\Delta V_{\text{th}} = V_{\text{th}}(\text{erase}) - V_{\text{th}}(\text{program})$ so that the sign convention matches ΔV_{hyst} , the transistor window falls from 0.581 V pristine to 0.243 V at 10^2 , 0.108 V at 10^3 , and -0.015 V at 10^4 [Fig. 8(b)]. The reversal is independent of the extraction criterion, being present at 1, 3, 10, 30 and 100 μA [Fig. 8(c)]. Both threshold states converge and cross, falling from 0.29/0.87 V pristine to 0.083/0.067 V at 10^4 .

The relation between ΔV_{hyst} and the pulse-written threshold window, stated as an assumption in Section II-D, is therefore supported by direct measurement: the two quantities collapse on the same schedule and change sign at the same cycle count [Fig. 7(c)]. We note that the two are measured on very different timescales — 60 s for the transistor read against ~ 17 ms per point for C - V — so their agreement indicates that the responsible charge is stable over at least that range. The off-state current floor of $\sim 3 \times 10^{-7}$ A in these devices prevents extraction of the subthreshold swing.

3.6 Polarization and Window Are Decoupled

Fig. 7 places the measurements together and constitutes the central result. Normalising each device’s window to its own pristine value [Fig. 7(a)], all five collapse to between -0.26 and -0.42 of their initial magnitude at 10^4 cycles, while the reference $2P_r$ rises to 1.45 of its pristine value at 10^2 and remains above unity at 10^4 .

The window efficiency of (8), evaluated against the common reference polarization, falls from 15.9–16.5% pristine to -3.2 to -4.2% at 10^4 cycles for devices 2, 3, 6 and 7 [Fig. 7(b), Table 2]; device-1 follows the same trajectory from a lower starting value. The agreement among four independent devices at both ends of the range is closer than the device-to-device spread in the pristine window itself.

3.7 Interpretation

At 10^4 cycles the polarization is present and larger than pristine, and both the capacitance and transistor windows have reversed sign. Over the range where the series correction is valid, the gate-stack capacitance is unchanged to better than 1% while most of the collapse occurs. These observations are consistent with charge accumulating within the channel-side gate stack: such charge can screen the polarization from the channel while leaving both the polarization and the series capacitance intact. From (9), the 0.51 V swing in ΔV_{hyst} on device-1 corresponds to $\Delta N_{\text{eq}} \approx 1.3 \times 10^{13} \text{ cm}^{-2}$, a density readily accommodated at an $\text{HfO}_2/\text{SiO}_2/\text{Si}$ interface [32, 30], and comparable to trap densities extracted by charge pumping and low-frequency noise on nominally identical stacks [17, 24].

The sign reversal further shows that this charge does not merely neutralize the polarization’s influence. It exceeds it, and there is necessarily an intermediate cycle count at which the two contributions cancel exactly.

The present measurements do not determine where the compensating charge resides. They do not distinguish charge in bulk HZO, at the HZO/ SiO_2 interface, within SiO_2 , or at the SiO_2/Si interface, and we describe it only as effective charge within the channel-side gate stack. Nor do we claim that the flank asymmetry, imprint drift and conduction increase share a single origin. The mechanism is presented as consistent with charge trapping rather than as an identification of the responsible traps; transmission electron microscopy with electron energy-loss spectroscopy, or charge pumping across the cycling range, would be required to localize and quantify it directly.

Three limitations remain. The five devices are from one die, and the zero-crossing cycle count varies by a factor of five among them. Polarization and transfer characteristics were each measured on a single device (Section II-B), so the decoupling argument combines a five-device window measurement with single-device polarization and transistor references. And η depends on the convention chosen for the ideal window in (8); alternative denominators shift the absolute values but not the sign change or the trend.

4 Conclusion

The DC-sweep C - V hysteresis window of an HfO_2 FeFET does not merely close with cycling: it reverses sign. The reversal is reproduced on five independent devices with zero crossings between 5×10^2 and 2×10^3 cycles, survives a ten-fold increase in measurement integration time, is present at all seven measurement frequencies and four sweep amplitudes, and is confirmed by transfer characteristics in which ΔV_{th} changes sign at the same cycle count. Over the same range the remanent polarization stands above its pristine value, and where the lumped C_p - R_p model is valid the gate-stack capacitance is constant to better than 1% while 64% of the collapse occurs. Loss of switchable

polarization alone therefore cannot account for the early collapse of the hysteresis window in these devices, and interfacial charge that first compensates and then overcompensates the polarization field provides a consistent account.

Acknowledgment

The authors thank the Taiwan Semiconductor Research Institute and the Nano-Fabrication Facility, National Yang-Ming Chiao Tung University, and Prof. Min-Hung Lee for the measurement facility at NTU.

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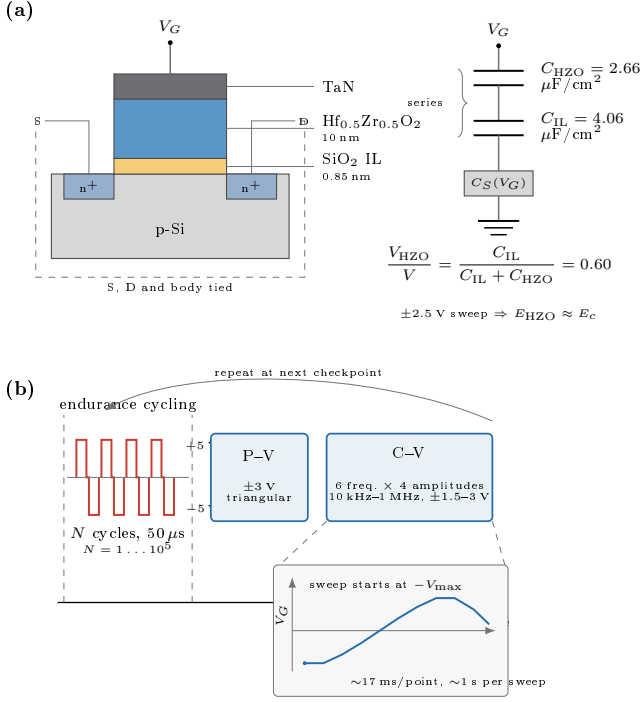


Figure 1: Device and measurement protocol. (a) Gate stack and equivalent series divider, (1); at most 60% of the gate bias reaches the ferroelectric, and with $V_c = 2.08$ V this gives $E_c \approx 1.25$ MV cm⁻¹. (b) At each checkpoint the device is cycled with ± 5 V, 50 μ s pulses, then characterized. The C - V sweep starts at the negative rail and takes ~ 17 ms per point, four orders of magnitude slower than the write pulse; the reported window is DC-sweep hysteresis, not a pulse-written state.

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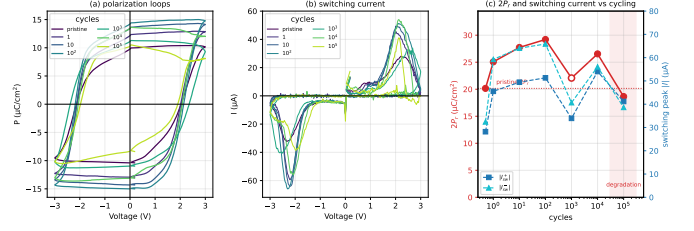


Figure 2: Ferroelectric response versus cycling. (a) P - V loops and (b) switching current. (c) $2P_r$ from (3) and switching peaks; dotted line marks the pristine $2P_r$. (d) V_{imp} and V_c from (4). Wake-up raises $2P_r$ 45% by 10^2 cycles and it stays above pristine through 10^4 , with V_c constant to 4%. At 10^5 , V_c decreases as the peaks broaden — not a domain-pinning signature. Measured on a single device of the die; used as the common polarization reference (Section II-B). Open symbol at 10^3 : single unverified trace.

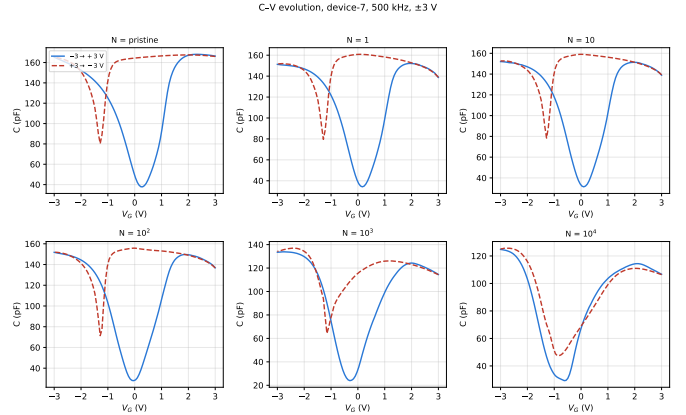


Figure 3: C - V evolution of device-7 at 500 kHz, ± 3 V. Solid: forward sweep; dashed: reverse. The forward-reverse separation narrows monotonically, closes near 10^3 cycles, and reopens at 10^4 with the branches exchanged.

Table 1: DC-Sweep Hysteresis Window Across Five Devices

N	dev-1	dev-2	dev-3	dev-6	dev-7
pristine	+0.363	+0.822	+0.804	+0.791	+0.803
1	+0.265	+0.920	+0.968	+0.719	+0.740
10	+0.221	+0.806	—	+0.606	+0.672
10^2	+0.122	+0.331	—	+0.396	+0.492
10^3	-0.052	-0.021	+0.129	+0.010	+0.172
10^4	-0.151	-0.265	-0.262	-0.205	-0.275
10^5	—	-0.115	-0.105	—	-0.116
zero cross.	503	871	2136	1117	2422

All values in volts, from (5). Devices 1–3 at 100 kHz, devices 6–7 at 500 kHz, all at ± 3 V. Entries omitted where the loss tangent exceeds the screening threshold.

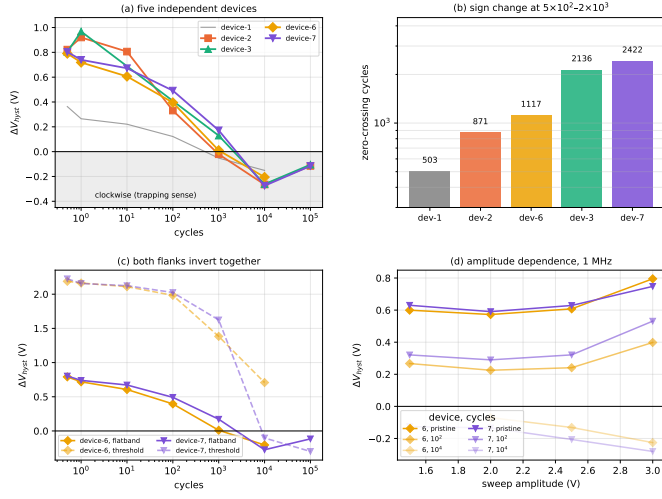


Figure 4: The window collapses and reverses sign. (a) ΔV_{hyst} versus cycling for five independent devices; shaded region marks clockwise loops. Device-1 (grey) has a smaller pristine window but the same trajectory. (b) Interpolated zero crossings: 503, 871, 1117, 2136 and 2422 cycles. (c) Flatband and threshold flanks, devices 6 and 7; both invert on the same schedule. (d) Window versus sweep amplitude at 1 MHz; the reversal is present at every amplitude.

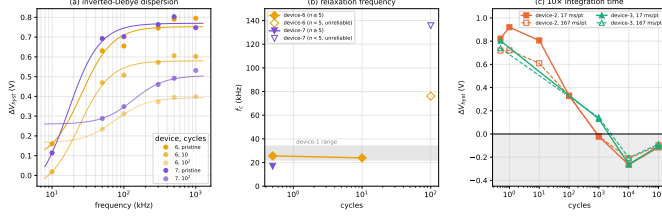


Figure 5: The reversal is not a measurement artifact. (a) Inverted-Debye dispersion, (10), devices 6 and 7. (b) Fitted f_c ; filled symbols are fits with ≥ 5 points, open symbols the four-point fits at 10² cycles which are not reliable. Grey band: device-1 range. (c) PLC 1 versus PLC 10 (16.7 versus 167 ms per point). The sign is unchanged at every checkpoint through 10⁵; magnitudes move toward zero at longer integration, opposite to the dependence expected from a leakage artifact.

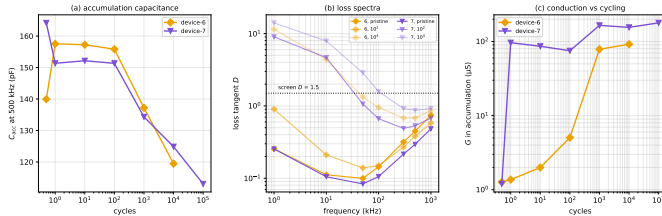


Figure 6: Gate-stack controls, devices 6 and 7. (a) Accumulation capacitance at 500 kHz. (b) Loss spectra; dotted line marks the screening threshold. (c) Accumulation conductance versus cycling.

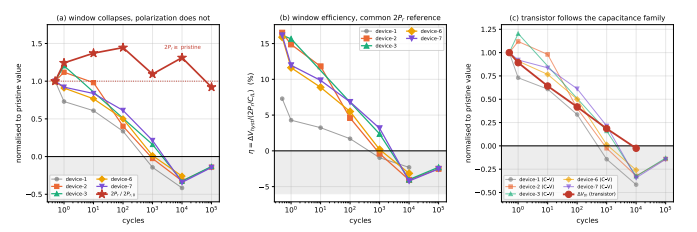


Figure 7: Polarization and window are decoupled. (a) ΔV_{hyst} normalised to each device's own pristine value, five devices, with $2P_r/2P_{r,0}$ from the common reference measurement. (b) Window efficiency (8) evaluated against that reference: 15.9–16.5% pristine falling to –3.2 to –4.2% at 10⁴ for devices 2, 3, 6 and 7. (c) Normalised ΔV_{th} (transistor, single device, 60s read delay) plotted against the normalised ΔV_{hyst} of all five devices.

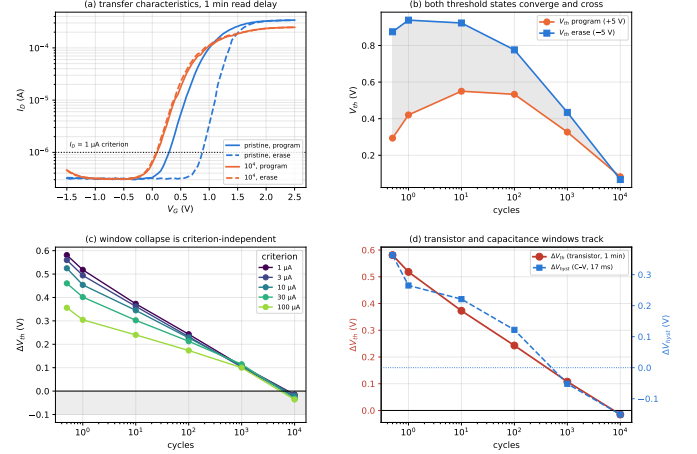


Figure 8: Transistor-level confirmation, single device. (a) Transfer characteristics after program and erase, pristine and 10⁴ cycles, 60s read delay; dotted line marks the 1 μ A extraction criterion. (b) V_{th} of both states, which converge and cross. (c) ΔV_{th} at five current criteria; the sign reversal is criterion-independent. (d) ΔV_{th} and ΔV_{hyst} versus cycling.

Table 2: Window Efficiency η (%) Against the Common Polarization Reference

N	$2P_r$ ($\mu C cm^{-2}$)	dev-1	dev-2	dev-3	dev-6	dev-7
pristine	20.2	7.3	16.5	16.2	15.9	16.1
1	25.1	4.3	14.9	15.7	11.6	12.0
10	27.7	3.2	11.8	—	8.9	9.9
10 ²	29.2	1.7	4.6	—	5.5	6.8
10 ³	22.1	–1.0	–0.4	2.4	0.2	3.2
10 ⁴	26.5	–2.3	–4.1	–4.0	–3.2	–4.2
10 ⁵	18.6	—	–2.5	–2.3	—	–2.5

η from (8). The $2P_r$ column is the common reference measured on a single device of the die (Section II-B) and is applied to all five devices; η therefore compares each device's window against a representative stack polarization rather than its own.

Table 3: Ferroelectric Switching Diagnostics

N	V_c (V)	E_c (MV cm ⁻¹)	V_{imp} (mV)	I_{sw}^+ (μ A)	I_{sw}^- (μ A)	I_{ns} (μ A)
pristine	2.08	1.25	-48	28.2	-32.4	3.15
1	2.07	1.24	-48	45.5	-59.3	1.46
10	2.10	1.26	-28	49.4	-64.1	0.83
10 ²	2.16	1.30	-23	51.3	-65.9	0.62
10 ³	2.37	1.42	-12	33.9	-40.7	1.52
10 ⁴	2.07	1.24	+9	54.0	-56.1	1.20
10 ⁵	1.94	1.16	-35	41.2	-38.6	1.87

From (4); $E_c = 0.60 V_c / t_{\text{HZO}}$ using (1). The 10³ row derives from a single unverified trace and is excluded from the quoted V_c variation.

Table 4: Transistor Threshold Window, Single Device

N	$V_{\text{th}}(\text{prog.})$ (V)	$V_{\text{th}}(\text{erase})$ (V)	ΔV_{th} (V)	$I_{\text{on}}/I_{\text{off}}$
pristine	0.294	0.875	+0.581	1053
1	0.421	0.939	+0.518	1050
10	0.550	0.923	+0.373	1063
10 ²	0.533	0.776	+0.243	1025
10 ³	0.327	0.434	+0.108	969
10 ⁴	0.083	0.067	-0.015	538

V_{th} at a constant current of 1 μ A; $\Delta V_{\text{th}} = V_{\text{th}}(\text{erase}) - V_{\text{th}}(\text{program})$ so that the sign convention matches ΔV_{hyst} . Read delay 60 s.